

Bruce Cheng

AI Application Engineer | Enterprise AI Architecture & LLM Systems
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EXPERIENCE

Information Engineer (Generative AI / LLM Projects)

Aug. 2025 – Present

PixArt Imaging Inc.

Hsinchu, Taiwan

- Own end-to-end delivery of on-prem GenAI solutions for Legal, R&D, and engineering teams under strict semiconductor security constraints.
- Designed a shared internal GenAI workspace around vLLM serving, document/image utilities, and task-oriented AI flows for engineering adoption.
- Delivered secure analytics and document-intelligence systems using guarded NL2SQL, multimodal review, evidence-grounded RAG, role-scoped access, and auditability.
- Productionized legal, patent, meeting-intelligence, and engineering automation workflows using local LLMs, OCR/ASR, Python-based tools, and validation logic.

AI Engineer / Project Leader

Aug. 2021 – Aug. 2025

AUO Corporation - Advanced Manufacturing Center

Taichung, Taiwan

- Led AI projects connecting models, data pipelines, explainability, and internal tools to manufacturing workflows.
- Improved yield by 1.43%, reduced defects by 12.62%, and saved approx. 300 man-hours monthly through Golden Path optimization and LIME analysis.
- Evolved an intelligent assistant from RAG into a tool-using agent workflow with SQL/API access.
- Reduced preventive-maintenance prediction RMSE by 20% and saved approx. 90 man-hours monthly with LSTM and a custom trend-consistency objective.

Research Intern

Feb. 2019 – Jul. 2019

Academia Sinica - GIS Center

Taipei, Taiwan

- Built a web-based Taiwan address normalization workflow; awarded Best Student Paper at TGIS Conference.

PROJECTS

Work-Hour Review & Insights Platform | *NL2SQL, Data Integration, Query Engine, FastAPI* 2026

- Built manager-scoped work-hour review dashboards and evidence-grounded NL2SQL using legacy-data ETL, historical organization snapshots, isolated read-only query views, and audit logs.

Multimodal Technical Document Review | *VLM, MCP, Multimodal Pipeline, FastAPI* 2026

- Preserved technical datasheet structure while combining multimodal analysis with Web, CLI, and MCP interfaces for reusable engineering review.

LLM Gateway & Operations Platform | *LLM Gateway, Model Routing, Usage Metering* 2025 – 2026

- Built an internal LLM gateway that unified model access, API key management, routing, and usage metering, with centralized dashboards for usage, cost, and service operations.

Evidence-Grounded Technical Knowledge | *VLM, RAG, Vector Search, Knowledge Extraction* 2026

- Designed a VLM-assisted, human-reviewed RAG workflow with source-cited retrieval and exact page-level evidence traceability.

TECHNICAL SKILLS

Generative AI & LLM Systems: On-Prem LLM Serving, vLLM, RAG, Agent & Tool-Calling Workflows, Multimodal / VLM, OCR / ASR, Prompt & Validation Design

AI System & Solution Design: Enterprise AI Architecture, Workflow Orchestration, API / Tool Integration, Security & Reliability Design

Backend & Data Engineering: Python, FastAPI, REST APIs, SQL, Docker, ETL / Data Pipelines, Databricks

Applied Machine Learning: Time-Series Forecasting, LSTM, Explainable AI, Optimization

EDUCATION

National Yunlin University of Science and Technology

Yunlin, Taiwan

Master of Science in Industrial Engineering and Management

2019 – 2021

National Yunlin University of Science and Technology

Yunlin, Taiwan

Bachelor of Science in Industrial Engineering and Management

2015 – 2019